

SCC.160 – Fundamentals of Communication Systems

Tutorial Questions and Revision (Reliable Transmission of Information)

Knowledge

Q1. How does a single-bit error differ from a burst error?

In a single-bit error only one bit of a data unit is corrupted; in a burst error more than one bit is corrupted (not necessarily contiguous).

Q2. Discuss the concept of redundancy in error detection and correction.

Redundancy is a technique of adding extra bits to each data unit to determine the accuracy of transmission.

Q3. Distinguish between forward error correction versus error correction by retransmission.

In forward error correction, the receiver tries to correct the corrupted codeword; in error detection by retransmission, the corrupted message is discarded (the sender needs to retransmit the message).

Q4. What is the definition of a linear block code? What is the definition of a cyclic code?

A linear block code is a block code in which the exclusive-or of any two codewords results in another codeword. A cyclic code is a linear block code in which the rotation of any codeword results in another codeword.

Q5. What is the Hamming distance? What is the minimum Hamming distance?

The Hamming distance between two words (of the same size) is the number of differences between the corresponding bits. The Hamming distance can easily be found if we apply the XOR operation on the two words and count the number of 1s in the result. The minimum Hamming distance is the smallest Hamming distance between all possible pairs in a set of words.

Q6. How is the simple parity check related to the two-dimensional parity check?

The single parity check uses one redundant bit for the whole data unit. In a two dimensional parity check, original data bits are organized in a table of rows and columns. The parity bit is then calculated for each column and each row.

Q7. What kind of arithmetic is used to add data items in checksum calculation?

One's complement arithmetic is used to add data items in checksum calculation. In this arithmetic, when a number needs more than n bits, the extra bits are wrapped and added to the number. In this arithmetic, the complement of a number is made by inverting all bits.

Q8. What kind of error is undetectable by the checksum?

At least three types of error cannot be detected by the current checksum calculation. First, if two data items are swapped during transmission, the sum and the checksum values will not change. Second, if the value of one data item is increased (intentionally or maliciously) and the value of another one is decreased (intentionally or maliciously) the same amount, the sum and the checksum cannot detect these changes. Third, if one or more data items is changed in such a way that the change is a multiple of $2^{16} - 1$, the sum or the checksum cannot detect the changes.

Q9. List the **three main operations** that can be performed on a message (e.g., a file, an image, a document, etc). How do they affect the size of the message?

A message can be encoded (its size increases) and be made more robust to channel noise. It can be compressed (its size drops) to reduce its space requirements. It can also be encrypted to make its content more secure from eavesdroppers, in which case its size can either be reduced, increased or remain the same, depending on the applied encryption rules.

Q10. How do **channel coding** and **modulation** affect data reliability and data rate (throughput)?

Channel coding increases data reliability but reduces data rate. High-order modulation can increase data rate at the expense of data reliability. Combining modulation and coding schemes allows us to adjust their trade-offs depending on the application.

Understanding

Q1. What is the maximum effect of a 2-ms burst of noise on data transmitted at the following rates?

- a. 1500 bps
- b. 12 kbps
- c. 100 kbps
- d. 100 Mbps

a. vulnerable bits = $(1,500) \times (2 \times 10^{-3}) = 3 \text{ bits}$

b. vulnerable bits = $(12 \times 10^3) \times (2 \times 10^{-3}) = 24 \text{ bits}$

c. vulnerable bits = $(100 \times 10^3) \times (2 \times 10^{-3}) = 200 \text{ bits}$

d. vulnerable bits = $(100 \times 10^6) \times (2 \times 10^{-3}) = 200,000 \text{ bits}$

Q2. Exclusive-OR (XOR) is one of the most used operations in the calculation of codewords. Apply the exclusive-OR operation on the following pairs of patterns. Interpret the results.

- a. $(10001) \oplus (10001)$
- b. $(11100) \oplus (00000)$
- c. $(10011) \oplus (11111)$

a. $(10001) \oplus (10001) = 00000$

b. $(11100) \oplus (00000) = 11100$

c. $(10011) \oplus (11111) = 01100$

Comment: The above shows three properties of the exclusive-or operation. First, the result of XORing two equal patterns is an all-zero pattern (part a). Second, the result of XORing of any pattern with an all-zero pattern is the original non-zero pattern (part b). Third, the result of XORing of any pattern with an all-one pattern is the complement of the original non-one pattern.

Q3. Using the code in Table below, what is the dataword if each of the following codewords is received?

- a. 01011
- b. 11111
- c. 00000
- d. 11011

Dataword	Codeword	Dataword	Codeword
0000	0000 0	1000	1 000 1
0001	000 1 1	1001	1 00 1 0
0010	00 1 0 1	1010	1 0 1 0 0
0011	00 1 1 0	1011	1 0 1 1 1
0100	0 1 00 1	1100	1 100 0
0101	0 1 0 1 0	1101	1 10 1 1
0110	0 1 10 0	1110	1 110 1
0111	0 1 11 1	1111	1 111 0

- a. error b. error c. 0000 d. 1101

Q4. What is the Hamming distance for each of the following codewords?

- $d(10000, 00000)$
- $d(10101, 10000)$
- $d(00000, 11111)$
- $d(00000, 00000)$

The following shows the result. Part d shows that the Hamming distance between a word and itself is 0.

- $d(10000, 00000) = 1$
- $d(10101, 10000) = 2$
- $d(00000, 11111) = 5$
- $d(00000, 00000) = 0$

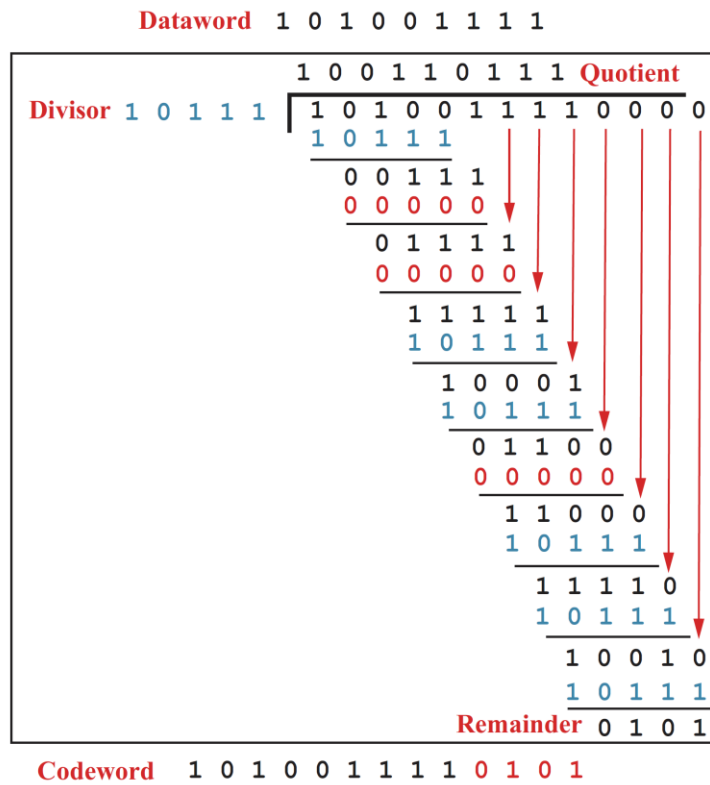
Q5. Assuming even parity, find the parity bit for each of the following data units.

- 1001011
- 0001100
- 1000000
- 1110111

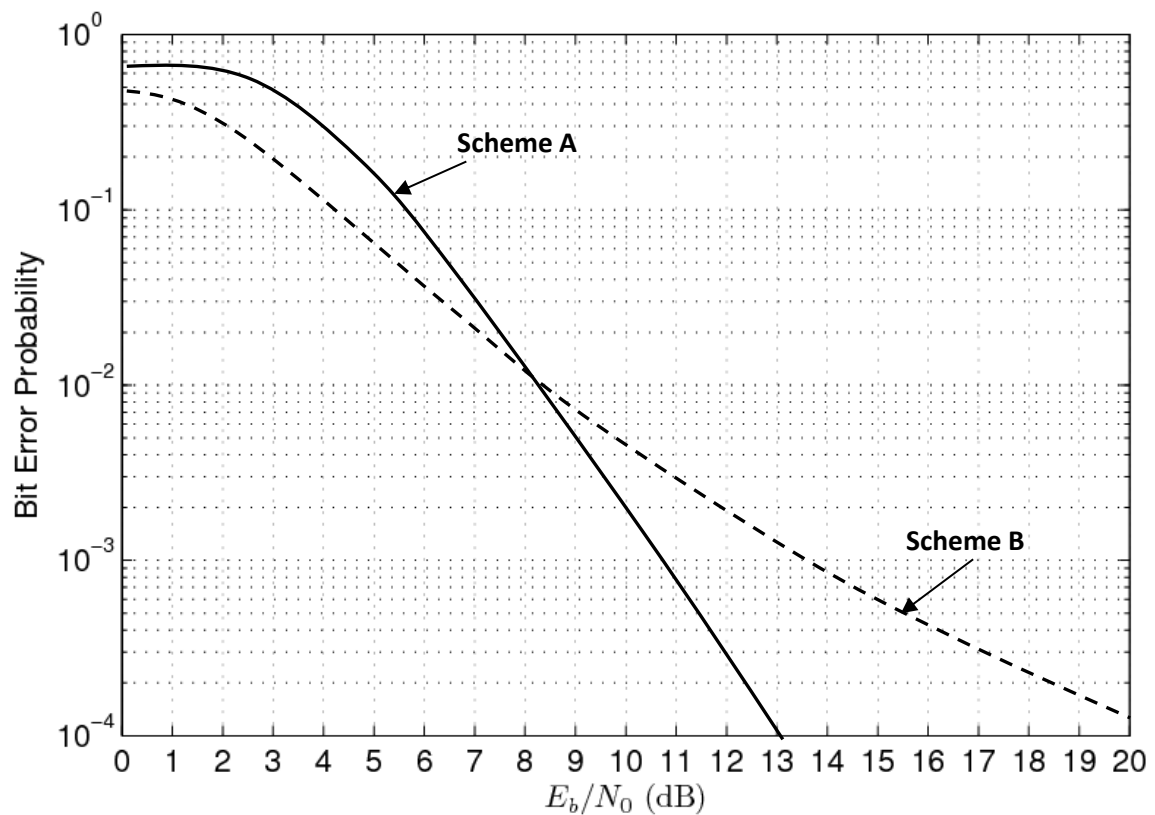
We need to add all bits modulo-2 (XORing). However, it is simpler to count the number of 1s and make them even by adding a 0 or a 1. We have shown the parity bit in the codeword in colour and separated for emphasis.

Dataword	Number of 1s	Parity	Codeword
a. 1001011 →	4 (even)	→ 0	100101 1 0
b. 0001100 →	2 (even)	→ 0	000110 0 0
c. 1000000 →	1 (odd)	→ 1	100000 1 1
d. 1110111 →	6 (even)	→ 0	111011 1 0

Q6. Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division).



Q7. The performance of **two** transmission schemes (A and B) is depicted in the figure below. Answer the following questions:



a. Which scheme is **better**?

Scheme B is better for low E_b/N_0 values (up to 8 dB), whilst Scheme A is better for E_b/N_0 values greater than 8 dB.

- b. If we transmit 5000 bits using Scheme B at $E_b/N_0 = 15$ dB, how many **bits** should we expect to receive in **error**?

From the plot, we see that the bit error probability of Scheme B at $E_b/N_0 = 15$ dB is 6×10^{-4} . Therefore, when we transmit 5000 bits, we expect $5000 \times 6 \times 10^{-4} = 3$ bits to be received in error.

- c. In order to achieve a bit error probability of 5×10^{-3} using Scheme A, how **many times** should the energy per bit E_b be higher than the power spectral density N_0 ?

A bit error probability of 5×10^{-3} can be achieved for $E_b/N_0 = 9$ dB when Scheme A is used. If we convert the value of 9 dB to a unitless number, i.e., $10^{(9\text{dB}/10)} = 7.94$, we find that E_b should be about 8 times higher than N_0 .

- d. Assume that Scheme A corresponds to an **uncoded modulation** scheme that uses **8 symbols**. What is the required **SNR** (in dB) to achieve a bit error probability of 10^{-4} ?

When Scheme A is employed, a bit error probability of 10^{-4} can be achieved for $E_b/N_0 = 13$ dB. If Scheme A has $8 = 2^3$ symbols available, we understand that 3 bits form a symbol. Therefore, the required SNR is $13 \text{ dB} + 10\log_{10}(3) = 17.77 \text{ dB}$.